National
Qualifications
2018

2018 Mathematics

National 5 - Paper 1

Finalised Marking Instructions

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General marking principles for National 5 Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

- *generic scheme — this indicates why each mark is awarded*
- *illustrative scheme — this covers methods which are commonly seen throughout the marking*

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.

(c) One mark is available for each •. There are no half marks.

(d) If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.

(e) Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.

(f) Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.

(g) If an error is trivial, casual or insignificant, for example 6 x 6 = 12, candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.

(h) If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$
$$x - 4x + 3 = 0$$
$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$
$$x - 4x + 3 = 0$$
$$(x-3)(x-1) = 0$$
$$x = 1 \text{ or } 3$$

(i) Horizontal/vertical marking

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

	$\bullet^5$	$\bullet^6$
$\bullet^5$	$x = 2$	$x = -4$
$\bullet^6$	$y = 5$	$y = -7$

Horizontal: $\bullet^5$ $x = 2$ and $x = -4$ Vertical: $\bullet^5$ $x = 2$ and $y = 5$

$\bullet^6$ $y = 5$ and $y = -7$ $\bullet^6$ $x = -4$ and $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ must be simplified to 43

$\frac{15}{0{\cdot}3}$ must be simplified to 50 $\frac{\frac{4}{5}}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8*

*The square root of perfect squares up to and including 100 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

  $(x^3 + 2x^2 + 3x + 2)(2x + 1)$ written as

  $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$

  $= 2x^4 + 5x^3 + 8x^2 + 7x + 2$

  gains full credit

- repeated error within a question, but not between questions or papers

(m) In any 'Show that…' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

(n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

(o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.

(p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

Strategy 1 attempt 1 is worth 3 marks.	Strategy 2 attempt 1 is worth 1 mark.
Strategy 1 attempt 2 is worth 4 marks.	Strategy 2 attempt 2 is worth 5 marks.
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

Detailed marking instructions for each question

Question	Generic scheme	Illustrative scheme	Max mark
1.	•[1] identify common denominator	•[1] $2\frac{\cdots}{15}+\frac{\cdots}{15}$ **or** $\frac{\cdots}{15}+\frac{\cdots}{15}$	**2**
	•[2] answer	•[2] $3\frac{2}{15}$ or $\frac{47}{15}$	

Notes:

1. Correct answer without working award 0/2
2. Do not penalise incorrect conversion of $\frac{47}{15}$ to a mixed number

Commonly Observed Responses:

Question	Generic scheme	Illustrative scheme	Max mark
2.	•[1] start expansion	•[1] $3x^2-3x+x-1$ **or** $2x^2-10$	**3**
	•[2] complete expansion	•[2] $3x^2-3x+x-1+2x^2-10$	
	•[3] collect like terms (see Note 2)	•[3] $5x^2-2x-11$	

Notes:

1. Correct answer without working award 3/3
2. Terms in the expression to be simplified must include a constant and two different powers of x eg
   (a) $3x^3-3x+x-1+2x^2-10=3x^3+2x^2-2x-11$ award 2/3 ✓×✓
   (b) $3(x-1)+1(x-1)+2(x^2-5)=3x-3+x-1+2x^2-10=2x^2+4x-14$ award 2/3 ✓×✓
   (c) $3x^2+1-1+2x^2-10=5x^2-10$ award 1/3 ✓××
3. For subsequent incorrect working, the final mark is not available

Commonly Observed Responses:

Question	Generic scheme	Illustrative scheme	Max mark
3.	•1 evidence of scaling (match x or y coefficients)	•1 eg $8x+10y=-6$, $30x-10y=25$	**3**
	•2 follow a valid strategy through to produce values for x and y	•2 values for x and y	
	•3 calculate correct values for x and y	•3 $x=0\cdot5,\ y=-1$	

Notes:

1. Correct answer without working — award 0/3
2. Answer obtained by guess and check — award 0/3

Commonly Observed Responses:

Question	Generic scheme	Illustrative scheme	Max mark
4.	•[1] evidence of subtraction	•[1] eg $\begin{pmatrix}6\\-4\\3\end{pmatrix}-\begin{pmatrix}1\\5\\1\end{pmatrix}$	**2**
	•[2] all components correct	•[2] $\begin{pmatrix}5\\-9\\2\end{pmatrix}$	

Notes:

1. Correct answer without working award 2/2

2. The maximum mark available is 1/2 where
   (a) brackets are omitted from the final answer
   (b) final answer is given in coordinate form

2. Accept one correct component for evidence of subtraction
   eg $\begin{pmatrix}5\\ \\ \end{pmatrix}$ or $\begin{pmatrix}\\-9\\ \end{pmatrix}$ or $\begin{pmatrix}\\ \\2\end{pmatrix}$ award 1/2 ✓×

Commonly Observed Responses:

1. (a) $\begin{pmatrix}1\\5\\1\end{pmatrix}+\begin{pmatrix}5\\-9\\2\end{pmatrix}=\begin{pmatrix}6\\-4\\3\end{pmatrix}$ award 2/2

   (b) $\begin{pmatrix}1\\5\\1\end{pmatrix}+\begin{pmatrix}5\\-9\\2\end{pmatrix}=\begin{pmatrix}6\\-4\\3\end{pmatrix}\rightarrow \mathbf{v}=\begin{pmatrix}6\\-4\\3\end{pmatrix}$ award 1/2

2. $\begin{pmatrix}-5\\9\\-2\end{pmatrix}$ $[\mathbf{u}-(\mathbf{u}+\mathbf{v})]$ no working necessary award 1/2 ×✓

3. $\begin{pmatrix}7\\1\\4\end{pmatrix}$ $[\mathbf{u}+(\mathbf{u}+\mathbf{v})]$ no working necessary award 1/2 ×✓

4. $(5,-9,2)$ or $5,-9,2$ award 1/2 ✓×

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Question			Generic scheme	Illustrative scheme	Max mark
5.			•[1] correct factorisation •[2] solve for x	•[1] $(x-3)(x-8)$ •[2] $(x=)3,(x=)8$	**2**

Notes:

1. Correct answer without working award 0/2
2. For an answer obtained by guess and check award 0/2
3. **BEWARE**
   •[2] is only available if 3,8 are clearly stated as solutions to $(x-3)(x-8)[=0]$ and not as factors of 24
4. Where quadratic formula is used award marks as follows
   •[1] $\frac{11\pm\sqrt{25}}{2}$
   •[2] 3,8

Commonly Observed Responses:

1. $(x+3)(x+8)\rightarrow 3,8$ award 0/2
2. $(x+4)(x-6)\rightarrow -4,6$ award 1/2 ×✓

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Question			Generic scheme	Illustrative scheme	Max mark
6.			•1 state the value of a •2 state the value of b	•1 $a = 5$ •2 $b = 4$	**2**

Notes:

Commonly Observed Responses:

1. (a) $y = 5\cos 4x$ award 2/2
   (b) $5, 4$ award 1/2

2. (a) $a = 4, b = 5$ or $y = 4\cos 5x$ award 1/2
   (b) $4, 5$ award 0/2

3. (a) $y = 5\cos 2x$ award 1/2
   (b) $5, 2$ award 0/2

Question			Generic scheme	Illustrative scheme	Max mark
7.	(a)		**Method 1:** $y-b=m(x-a)$ •1 calculate gradient •2 substitute gradient and a point into $y-b=m(x-a)$ •3 state equation in simplest form in terms of P and d **Method 2:** $y=mx+c$ •1 calculate gradient •2 substitute gradient and a point into $y=mx+c$ •3 state equation in simplest form in terms of P and d	**Method 1** •1 $\frac{6}{4}$ or equivalent •2 eg $y-20=\frac{6}{4}(x-12)$ •3 $P=\frac{3}{2}d+2$ or equivalent **Method 2** •1 $\frac{6}{4}$ or equivalent •2 eg $20=\frac{6}{4}\times 12+c$ •3 $P=\frac{3}{2}d+2$ or equivalent	**3**

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Question	Generic scheme	Illustrative scheme	Max mark

Notes:

1. Correct answer without working award 0/3
2. Gradient need not be simplified for the award of $\bullet^2$
3. Where $\frac{6}{4}$ is simplified incorrectly $\bullet^2$ is still available

   eg $m=\frac{6}{4}=\frac{2}{3} \rightarrow y-20=\frac{2}{3}(x-12) \rightarrow P=\frac{2}{3}d+12$ award 2/3 ✓✓×
4. $\bullet^3$ is not available where the calculated gradient is an integer
5. $\bullet^3$ is not available where a decimal **approximation** is used for the gradient

   eg $m=\frac{4}{6}=0\cdot67 \rightarrow y-20=0\cdot67(x-12) \rightarrow P=0\cdot67d+11\cdot96$ award 1/3 ×✓×
6. $\bullet^3$ is not available for invalid subsequent working

   eg $P=\frac{3}{2}d+2 \rightarrow 2P=3d+2$ award 2/3 ✓✓×

Commonly Observed Responses:
Working must be shown.

1. $y=\frac{3}{2}x+2$ award 2/3 ✓✓×
2. (a) $P=\frac{3}{2}d+2 \rightarrow 2P=3d+4$ award 3/3

   (b) $P=\frac{6}{4}d+2 \rightarrow 4P=6d+8$ award 2/3 ✓✓×
3. $m=\frac{6}{4}=0\cdot67 \rightarrow y-20=0\cdot67(x-12) \rightarrow P=0\cdot67d+11\cdot96$ award 2/3 ✓✓×
4. $m=\frac{6}{4}=\frac{3}{4}=0\cdot75 \rightarrow y-20=0\cdot75(x-12) \rightarrow P=0\cdot75d+11$ award 2/3 ✓✓×
5. $m=\frac{4}{6}=\frac{2}{3} \rightarrow y-20=\frac{2}{3}(x-12) \rightarrow P=\frac{2}{3}d+12$ award 2/3 ×✓✓
6. $m=\frac{3}{4}=0\cdot75 \rightarrow y-20=0\cdot75(x-12) \rightarrow P=0\cdot75d+11$ award 2/3 ×✓✓

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Question			Generic scheme	Illustrative scheme	Max mark
	(b)		•1 calculate cost	•1 $(£)\,9{\cdot}50$	1

Notes:

1. Correct answer without working award 1/1
2. Do not penalise omission of £
3. Do not accept $9{\cdot}5$ or $\frac{19}{2}$
4. Follow through mark from part (a) is only available if 5 is multiplied or divided by a non-unitary fraction (or decimal equivalent) followed by an addition or subtraction

Commonly Observed Responses:

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Question			Generic scheme	Illustrative scheme	Max mark
8.			•1 find discriminant •2 state nature of roots	•1 -24 •2 no real roots	**2**

Notes:

1. Correct answer without working award 0/2
2. $16-40<0 \rightarrow$ no real roots, award 2/2
3. Do **not** accept 'no roots', 'no real (or) distinct roots', 'no real (and) distinct roots'
4. Expected answer(s) for the award of •2, when
   (a) $b^2-4ac>0$: 'two real (and) distinct roots'
   (b) $b^2-4ac=0$: 'one repeated real root' or 'two equal real roots'

Commonly Observed Responses:

1. $\frac{-4\pm\sqrt{16-40}}{4}=\frac{-4\pm\sqrt{-24}}{4}$ award 1/2 ✓×
2. $\sqrt{16-40}=\sqrt{-24}$ award 1/2 ✓×

Question			Generic scheme	Illustrative scheme	Max mark
9.			•1 calculate the size of an interior angle of the decagon **or** angle JKL •2 calculate the size of angle KJL	•1 interior angle $=72+72$ **or** JKL $=36$ •2 127	**2**

Notes:

1. Correct answer without working award 2/2
2. Degree signs are not required
3. Full marks may be awarded for information marked on the diagram
4. For a **final** answer of 36 which is **not** named or shown at JKL award 0/2
5. Where JKL has been calculated incorrectly •2 is only available where there is **clear evidence** that JKL has been calculated by using the interior angle or exterior angle of the decagon

Commonly Observed Responses:

Question			Generic scheme	Illustrative scheme	Max mark
10.			•[1] correct substitution into cosine rule •[2] calculate XY^2 •[3] calculate XY	•[1] $10^2+8^2-2\times10\times8\times\frac{1}{8}$ •[2] 144 •[3] 12 (cm)	**3**

Notes:

1. Correct answer without working award 0/3
2. For $10^2+8^2-2\times10\times8\times\cancel{\cos}\frac{1}{8}=164-160\times\cancel{\cos}\frac{1}{8}=144\rightarrow12$
   where $\cos$ is scored out in **each line** of working award 3/3
3. •[3] is available for
   (a) expressing $\sqrt{x}$ as a surd in its simplest form
   eg $\sqrt{10^2+8^2}=\sqrt{164}=2\sqrt{41}$ award 1/3 ××✓
   [•[3] is not available where $\sqrt{x}$ cannot be simplified]
   (b) calculating $\sqrt{x}$ where x is a perfect square greater than 100
   eg $\sqrt{10^2-8^2}=\sqrt{36}=6$ award 0/3

Commonly Observed Responses:

1. $10^2+8^2-2\times10\times8\times\cos\frac{1}{8}=12$ award 2/3 ×✓✓
2. $10^2+8^2-2\times10\times8\times\cos\frac{1}{8}$ award 0/3

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Question			Generic scheme	Illustrative scheme	Max mark
11.			•[1] express as equivalent fraction with rational denominator •[2] express in simplest form	•[1] $\frac{9\sqrt{6}}{6}$ •[2] $\frac{3\sqrt{6}}{2}$	**2**

Notes:

1. Correct answer without working award 0/2

2. Accept $1{\cdot}5\sqrt{6}$

3. For subsequent incorrect working, the final mark is not available

   eg $\frac{9\sqrt{6}}{6} = \frac{3\sqrt{6}}{2} = 3\sqrt{3}$ award 1/2 ✓×

Commonly Observed Responses:

Question			Generic scheme	Illustrative scheme	Max mark
12.			$\bullet^1$ state value	$\bullet^1\ -0\cdot5$	**1**
Notes:					
Commonly Observed Responses:					

Question			Generic scheme	Illustrative scheme	Max mark
13.			$\bullet^1$ state coordinates of B $\bullet^2$ state coordinates of C	$\bullet^1\ (4,\ 8,\ 5)$ $\bullet^2\ (6,\ 8,\ 0)$	**2**

Notes:

1. The maximum mark available is 1/2 where
   (a) brackets are omitted (unless already penalised in Q4)
   (b) answers are given in component form (unless already penalised in Q4)

2. (a) For $(4,\ 8,\ 5)$ and $(6,\ 8,\ 0)$ award 2/2
   (b) For $C(4,\ 8,\ 5)$ and $B(6,\ 8,\ 0)$ award 1/2

3. For eg $(5,\ 4,\ 8)$ **and** $(0,\ 6,\ 8)$ [repeated error] award 1/2

4. $\bullet^2$ is available for answers of the form $B(x,\ y,\ z)$, $C(x+2,\ y,\ z-5)$
   eg $(2,\ 8,\ 5)$ and $(4,\ 8,\ 0)$

Commonly Observed Responses:

1. 4,8,5 and 6,8,0 award 1/2 ×✓
   (award 2/2 if omission of brackets has already been penalised in Q4)

2. $\begin{pmatrix}4\\8\\5\end{pmatrix}$ and $\begin{pmatrix}6\\8\\0\end{pmatrix}$ or $\begin{matrix}4\\8\\5\end{matrix}$ and $\begin{matrix}6\\8\\0\end{matrix}$ award 1/2 ×✓
   (award 2/2 if use of coordinates instead of components has already been penalised in Q4)

3. $\begin{pmatrix}5\\8\\4\end{pmatrix}$ and $\begin{pmatrix}6\\8\\0\end{pmatrix}$ award 0/2

Question			Generic scheme	Illustrative scheme	Max mark
14.			• [1] subtract h • [2] divide by g • [3] square	• [1] $y-h=g\sqrt{x}$ • [2] $\sqrt{x}=\dfrac{y-h}{g}$ • [3] $x=\left(\dfrac{y-h}{g}\right)^2$	**3**

Notes:

1. Correct answer without working award 1/3

2. For subsequent incorrect working, the final mark is not available

Commonly Observed Responses:
Working must be shown.

1. $x=\left(\dfrac{y}{g}-\dfrac{h}{g}\right)^2$ award 3/3

2. $x=\dfrac{y-h}{g}^{\ 2}$ award 2/3 ✓✓×

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Question			Generic scheme	Illustrative scheme	Max mark
15.			•1 start process •2 complete process	•1 $\frac{4}{9}$ **or** p^8 •2 $\frac{4}{9}p^8$	**2**

Notes:

1. For a correct answer without working award 2/2

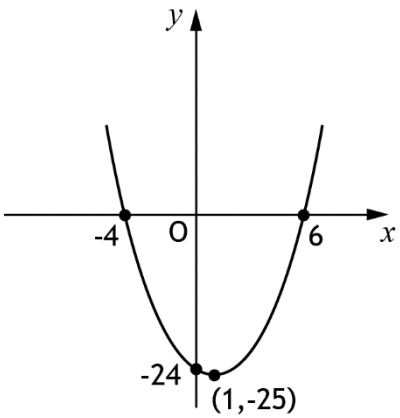
2. For subsequent incorrect working, the final mark is not available

3. **BEWARE** : For $\frac{2}{3}p^4+\frac{2}{3}p^4=\frac{4}{3}p^8$ award 0/2

4. Award •1 for an incorrect expansion leading to $\frac{4}{9}+\ldots$ **or** $\ldots+p^8$

   eg $\left(\frac{2}{3}p^4\right)\left(\frac{2}{3}p^4\right)=\frac{4}{9}+\frac{2}{3}p^4+\frac{2}{3}p^4+p^8=\frac{4}{9}+\frac{4}{3}p^4+p^8$ award 1/2 ✓×

Commonly Observed Responses:

Question			Generic scheme	Illustrative scheme	Max mark
16.			•[1] identify roots •[2] identify turning point **OR** y-intercept •[3] identify the turning point **AND** the y-intercept and sketch a consistently annotated parabola	•[1] -4 **AND** 6 •[2] $(1, -25)$ **OR** -24 •[3] $(1, -25)$ **AND** -24 and consistently annotated parabola (see Note 1).  	**3**

Notes:

1. •[3] is only available where the roots, turning point **AND** y-intercept are clearly marked and consistently annotated on the sketch

2. Accept correctly calculated roots and/or y-intercept annotated as $(0, -4)$, $(0, 6)$ and $(-24, 0)$ as evidence for the award of •[3] (treat as bad form)

3. •[3] is not available if graph is not a parabola
   eg roots $= -6 \text{ and } 4 \rightarrow (-1, -21)$ or -24 award 1/3 ×✓×

Commonly Observed Responses:

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Question			Generic scheme	Illustrative scheme	Max mark
17.			**Method 1**	**Method 1**	**3**
			•[1] start substitution into formula	•[1] $\frac{1}{3}\times 6^2\times h$ or $\frac{1}{3}Ah=138$	
			•[2] complete substitution into formula	•[2] $\frac{1}{3}\times 6^2\times h=138$	
			•[3] calculate height	•[3] 11·5 (cm)	
			Method 2	**Method 2**	
			•[1] change subject to h	•[1] $\frac{3V}{A}$ or $\frac{V}{\frac{1}{3}A}$	
			•[2] correct substitution into formula	•[2] $\frac{3\times 138}{6\times 6}$ or $\frac{138}{\frac{1}{3}\times 6\times 6}$	
			•[3] calculate height	•[3] 11·5 (cm)	

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Question	Generic scheme	Illustrative scheme	Max mark

Notes:

1. For a correct answer without working award 0/3
2. For an answer obtained by guess and check award 0/3
3. Accept $\frac{23}{2}$ or $11\frac{1}{2}$
4. For subsequent incorrect working, the final mark is not available
   eg $\frac{23}{2} = 11{\cdot}1$
5. Calculation must involve division by a number greater than 10 for the award of $\bullet^3$

Commonly Observed Responses:
Working must be shown.

1. (a) $\frac{1}{3}\times 6^2 \times h = 138 \rightarrow \frac{1}{3}\times 12 \times h = 138 \rightarrow 4h = 138 \rightarrow h = 34{\cdot}5$ award 2/3 ✓✓×

   (b) $\frac{1}{3}\times 12 \times h = 138 \rightarrow 4h = 138 \rightarrow h = 34{\cdot}5$ award 1/3 ✓××

2. $\frac{1}{3}\times 18 \times h = 138 \rightarrow 6h = 138 \rightarrow h = 23$ $[A = \frac{1}{2}\times 6^2]$ award 1/3 ✓××

3. $\frac{1}{3}\times 24 \times h = 138 \rightarrow 8h = 138 \rightarrow h = 17{\cdot}25$ $[A = 4\times 6]$ award 1/3 ✓××

4. (a) $V = \frac{1}{3}Ah \rightarrow \frac{1}{3}\times 6 \times h = 138 \rightarrow 2h = 138 \rightarrow h = 69$ award 1/3 ✓××

   (b) $h = \frac{3V}{A} \rightarrow h = \frac{3\times 138}{6} \rightarrow h = 69$ award 1/3 ✓××

   (c) $\frac{1}{3}\times 6 \times h = 138 \rightarrow 2h = 138 \rightarrow h = 69$ award 0/3

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Question	Generic scheme	Illustrative scheme	Max mark
18.	•¹ correct substitution for $\tan x$ •² express in simplest form	•¹ $\sin x \cos x \frac{\sin x}{\cos x}$ •² $\sin^2 x$	**2**

Notes:

1. For $\sin^2 x$ without working award 0/2
2. Degree signs are not required
3. Accept $(\sin x)^2$ but **not** $\sin x^2$
4. •¹ is not available if there are no variables
   eg $\sin\cos\frac{\sin}{\cos} = \sin^2$ award 1/2 ×✓
5. •¹ is not available if candidate simply states $\tan x = \frac{\sin x}{\cos x}$ then proceeds no further
6. •² is not available if there is invalid subsequent working
   eg (a) $\sin x \cos x \frac{\sin x}{\cos x} = \sin^2 x = 1 - \cos x$ award 1/2 ✓×
   (b) $\sin x \cos x \frac{\sin x}{\cos x} = \sin^2 x = 1 - \cos^2 x$ award 2/2
7. Alternative acceptable strategy:
   •¹ $\left(\frac{o}{h}\right)\left(\frac{a}{h}\right)\left(\frac{o}{a}\right)$
   •² $\frac{o^2}{h^2} = \sin^2 x$ award 2/2

Commonly Observed Responses:

1. $\sin x \cos x \frac{\sin x}{\cos x} = \sin x^2$ award 1/2 ✓×
2. $1 \times \frac{\sin x}{\cos x} = \tan x$ award 1/2 ✓×
3. $\sin x \cos x \frac{\cos x}{\sin x} = \cos^2 x$ award 1/2 ×✓

Question			Generic scheme	Illustrative scheme	Max mark
19.	(a)	(i)	•[1] correct bracket with square •[2] complete process	•[1] $(x-3)^2\ldots$ •[2] $(x-3)^2-90$	**2**

Notes:

1. Correct answer without working award 2/2
2. If the solution to (a)(i) only appears in (a)(ii) then both marks are available

Commonly Observed Responses:
No working necessary:

1. Award 2/2 for (a) $(x-3)^2+(-90)$ or $(x-3)^2+-90$
   (b) $(x-3)(x-3)-90$

2. Award 1/2 ×✓ for (a) $(x-3)-90$
   (b) $(x^2-3)-90$
   (c) $(x^2-3)^2-90$
   (d) $(x-3x)^2-90$
   (e) $(x-6)^2-117$

3. (a) $(x-6)^2-117$ award 1/2 ×✓
   (b) $(x-6)^2-90$ award 0/2

		(ii)	•[1] state equation of axis of symmetry	•[1] $x=3$	**1**

Notes:

1. For 3 or axis of symmetry $=3$ award 0/1
2. Answer must be consistent with answer to 19(a)(i) [unless no answer given to 19(a)(i)]

Commonly Observed Responses:

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Question			Generic scheme	Illustrative scheme	Max mark
19.	(b)		**Ans:** $d=3, e=10$		**4**
			Method 1	**Method 1**	
			$\bullet^1$ equate complete square form to zero	$\bullet^1$ $(x-3)^2-90=0$	
			$\bullet^2$ start to solve	$\bullet^2$ $x-3=\pm\sqrt{90}$	
			$\bullet^3$ solve equation	$\bullet^3$ $x=3\pm\sqrt{90}$	
			$\bullet^4$ complete process	$\bullet^4$ $d=3, e=10$ or $3\pm3\sqrt{10}$	
			Method 2	**Method 2**	
			$\bullet^1$ correct substitution into quadratic formula	$\bullet^1$ $\dfrac{6\pm\sqrt{(-6)^2-4\times1\times(-81)}}{2\times1}$	
			$\bullet^2$ evaluate discriminant	$\bullet^2$ 360 (stated or implied by $\bullet^3$)	
			$\bullet^3$ express discriminant in simplest form	$\bullet^3$ $6\sqrt{10}$	
			$\bullet^4$ complete process	$\bullet^4$ $d=3, e=10$ or $3\pm3\sqrt{10}$	

Notes:

1. Correct answer without working award 0/4
2. Where a, b and c are all positive $\bullet^2$ is not available
3. Where b^2-4ac is calculated incorrectly, $\bullet^3$ and $\bullet^4$ are only available if $b^2-4ac>0$ (See CORs 2 – 5)
4. $\bullet^4$ is only available where a correct simplification of $\sqrt{\text{discriminant}}$ leads to a final answer of the form $d\pm d\sqrt{e}$ (See COR 6)

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Commonly Observed Responses:
Working must be shown.

1. $\frac{6\pm\sqrt{(-6)^2-4\times1\times(-81)}}{2\times1}\rightarrow\frac{6\pm\sqrt{360}}{2}\rightarrow\frac{6\pm3\sqrt{40}}{2}\rightarrow3\pm1{\cdot}5\sqrt{40}$ award 2/4 ✓✓××

2. $\frac{6\pm\sqrt{(-6)^2-4\times1\times(-81)}}{2\times1}\rightarrow\frac{6\pm\sqrt{-288}}{2}\rightarrow\frac{6\pm12\sqrt{2}}{2}\rightarrow3\pm6\sqrt{2}$ award 1/4 ✓×××

   (**BEWARE:** candidate may get $\sqrt{-288}$ then change it to $\sqrt{288}$)

3. $\frac{6\pm\sqrt{(-6)^2-4\times1\times(-81)}}{2\times1}\rightarrow\frac{6\pm\sqrt{288}}{2}\rightarrow\frac{6\pm12\sqrt{2}}{2}\rightarrow3\pm6\sqrt{2}$ award 2/4 ✓×✓×

4. $\frac{6\pm\sqrt{(-6)^2-4\times1\times81}}{2\times1}\rightarrow\frac{6\pm\sqrt{-288}}{2}\rightarrow\frac{6\pm12\sqrt{2}}{2}\rightarrow3\pm6\sqrt{2}$ award 1/4 ×✓××

   (**BEWARE:** candidate may get $\sqrt{-288}$ then change it to $\sqrt{288}$)

5. $\frac{-6\pm\sqrt{6^2-4\times1\times81}}{2\times1}\rightarrow\frac{-6\pm\sqrt{-288}}{2}\rightarrow\frac{-6\pm12\sqrt{2}}{2}\rightarrow-3\pm6\sqrt{2}$ award 0/4

   (**BEWARE:** candidate may get $\sqrt{-288}$ then change it to $\sqrt{288}$)

6. $\frac{6\pm\sqrt{(-6)^2-4\times1\times81}}{2\times1}\rightarrow\frac{6\pm\sqrt{288}}{2}\rightarrow\frac{6\pm12\sqrt{2}}{2}\rightarrow3\pm6\sqrt{2}$ award 1/4 ××✓×

7. $\frac{6\pm\sqrt{(-6)^2-4\times1\times(-81)}}{2\times1}\rightarrow\frac{6\pm\sqrt{288}}{2}\rightarrow\frac{6\pm6\sqrt{8}}{2}\rightarrow3\pm3\sqrt{8}$ award 2/4 ✓××✓

8. $6\pm\sqrt{(-6)^2-4\times1\times(-81)}\rightarrow6\pm\sqrt{360}\rightarrow6\pm6\sqrt{10}$ award 2/4 ×✓✓×

[END OF MARKING INSTRUCTIONS]